Project 1: Hydraulic System for Waslala Nicaragua

MECH 3100: Fluid Mechanics

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Problem overview

The objective is to design the hydraulic system of a new school building in Waslala, Nicaragua. The new water system will deliver potable water from the existing city municipal supply line to the school building. The project evaluates the current available pressure and flow rate at the closest available water line, determining the required pump head, and selecting the proper pipe material and fittings to create a reliable and sustainable hydraulic system. While the technical goal is to calculate the flow distribution, determine losses and pump requirements, the broader purpose of this design is to apply engineering design principles to create access to clean water improving the community's infrastructure and increasing social well being. In a rural location where infrastructure may be unreliable and lacking in regular maintenance, a reliable hydraulic system will improve student well being by improving sanitation, reducing disease transmission, and creating a healthier learning environment for students and teachers. By minimizing major and minor losses and choosing efficient pipe and pump components, the system will minimize energy consumption which will decrease the school's environmental footprint by reducing unnecessary maintenance and material waste. Economically, a school in a lower-resource rural area must prioritize long term-affordability, so an efficient hydraulic design with lower operating costs and reduced energy usage is essential. Engineering design choices influence sustainable developments, such as improving access to reliable and safe infrastructure.

Design

Figure 1 is the designed hydraulic system; it consists of a 4 inch diameter main pipe from the city connection, a 4 inch vertical riser, two 2 inch horizontal branches which lead to outlets 1 and 2 at elevations z1 and z2, and a 4 inch diameter branch that leads to outlet 3 at elevation z3. Red indicates a tee pipe (4x2x4) connection and gray indicates an elbow (long radius 90° flanged); all fittings are flanged to promote lower turbulence for better flow. Gate and globe valves were used in our design, indicated by an x surrounded by a square or circle, depending on which type of valve is used.

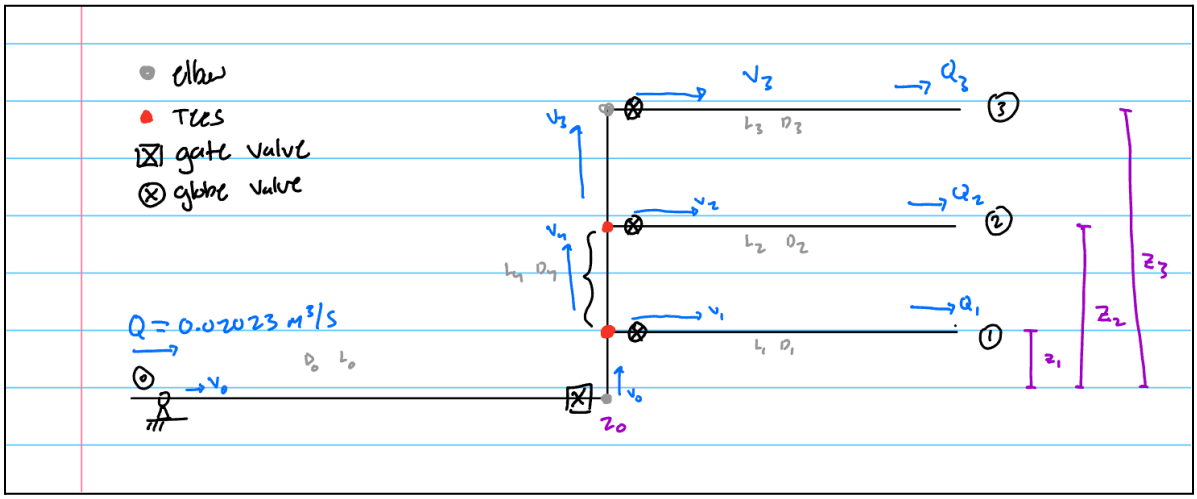


Figure 1: Sketch of Piping Route

Using the given flow rate of the closest water line of 320.65 gallons per minute at a pressure of 15 kPa it was determined to be inadequate to deliver water to the school. Detailed calculations are found in the Appendix, Figure 1A. The available pressure head at the school connection was computed as 1.529 m (5.02 ft). Including the inlet velocity head, the total available head is only 1.846 m (6.05 ft). The highest elevation is located 12m above the inlet, this 12m only represents the elevation gain not including losses. The Appendix calculations in Figure 1A shows that at least 12 m of head is required to have water flow to the highest outlet, this is without accounting

for any major or minor losses. The municipal supply is only 1.847 m which is not enough, this indicates a pump is required. In the design, when all major losses, minor losses, and required outlet head are included in the MATLAB script the system needs approximately 23.3 m (76 ft) of pump head.

For the pipe design the chosen pipe material was Polyvinyl Chloride (PVC) and the pipe fittings selected were flanged. PVC was selected because it is cost effective, corrosion and chemically resistant, and easier to assemble because it is lightweight [1]. To account for environmental impact, PVC is also recyclable and its long lifespan reduces the need for frequency replacement, which reduces waste [2]. PVC is considered more sustainable than other materials and due to the reduced friction from the smoothness of PVC and leak-free joints this also leads to greater water conservation [1].

Equivalent Roughness PVC (ε) : 0 (smooth), used copper Roughness 0.000002

The variable names on the sketch were used in the MATLAB code as shown below:

Lengths of pipes 0-4: L_0 , L_1 , L_2 , L_3 , L_4

Diameters of pipes 0-4: D_0 , D_1 , D_2 , D_3 , D_4

Elevations: z_0 , z_1 , z_2, z_3

Total Flow from the city line (given): Q_0

Flow rates in branches 1-3: Q_1 ,Q_2 Q_3

The following equations applied in the pipe design MATLAB script to solve for pump head (h_p) were used:

Conservation of mass:

$$V_0 A_0 = V_1 A_1 + V_4 A_4$$

$$V_4 A_4 = V_2 A_2 + V_3 A_3$$

Bernoulli from point 0 to point 1:

$$\frac{P_0}{\gamma} + \frac{V_0^2}{2g} + z_0 = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1 + f_0 \frac{L_0}{D_0}\frac{V_0^2}{2g} + f_1 \frac{L_1}{D_1}\frac{V_1^2}{2g} - h_p + h_{lm}$$

Bernoulli from point 0 to point 2:

$$\frac{P_0}{\gamma} + \frac{V_0^2}{2g} + z_0 = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1 + f_0 \frac{L_0}{D_0}\frac{V_0^2}{2g} + f_4 \frac{L_4}{D_4}\frac{V_4^2}{2g} + f_2 \frac{L_2}{D_2}\frac{V_2^2}{2g} - h_p + h_{lm}$$

Bernoulli from point 0 to point 3:

$$\frac{P_0}{\gamma} + \frac{V_0^2}{2g} + z_0 = \frac{P_1}{\gamma} + \frac{V_1^2}{2g} + z_1 + f_0 \frac{L_0}{D_0}\frac{V_0^2}{2g} + f_4 \frac{L_4}{D_4}\frac{V_4^2}{2g} + f_2 \frac{L_3}{D_3}\frac{V_3^2}{2g} - h_p + h_{lm}$$

Major losses, represented by $f \frac{L}{D}\frac{V^2}{2g}$, helps determine how much extra head the pump must supply so that each branch of the piping system satisfies Bernoulli's equation. The major losses shape the velocity distribution and final pump head. The minor losses represent turbulence from fittings and components and is represented in the code by $K\frac{V^2}{2g}$, where K is the loss coefficient. In the code, both major and minor losses are computed for each pipe and flow path, and added into the Bernoulli equations to determine pump head. Given the Bernoulli equations, the pump head must overcome elevation, required head outlet, all major losses, and all minor losses. Since the moody diagram was not used to find the friction factor in the MATLAB script, for turbulent flow the friction factor was calculated using the Colebrook-White equation as shown below:

$$\frac{1}{\sqrt{f}} = -2log\left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{Re\sqrt{f}}\right)$$

ε: Pipe Roughness

Re: Reynolds Number

f: friction coefficient

D: pipe diameter

The Colebrook-White equation is used to calculate the friction factor for turbulent flow in pipes. This friction factor is needed to determine pressure drop or head loss due to friction in a pipeline. The code uses “fzero” to solve, updating the friction factor based on the new Reynolds number and roughness.

Results

As printed in the “Final results” output on our final MATLAB script, given the provided flow rate of 320.65-gpm (0.02023 m^3/s); the final output resulted in a required pump head of 76.638 ft of water (23.359 m^3/s) for flow to reach the top branch of pipe 3.

```
FINAL RESULTS:
hp  = 23.359 m of water
Q0  = 0.02023 m^3/s
Q1 = 0.00891,  Q2 = 0.00649,  Q3 = 0.00483 m^3/s
Continuity: Q0 = 0.02023,  Q1+Q2+Q3 = 0.02023
Q0 = 320.65 gpm
Q1 = 141.26 gpm, Q2 = 102.84 gpm, Q3 = 76.55 gpm
hp = 76.638 ft of water
```

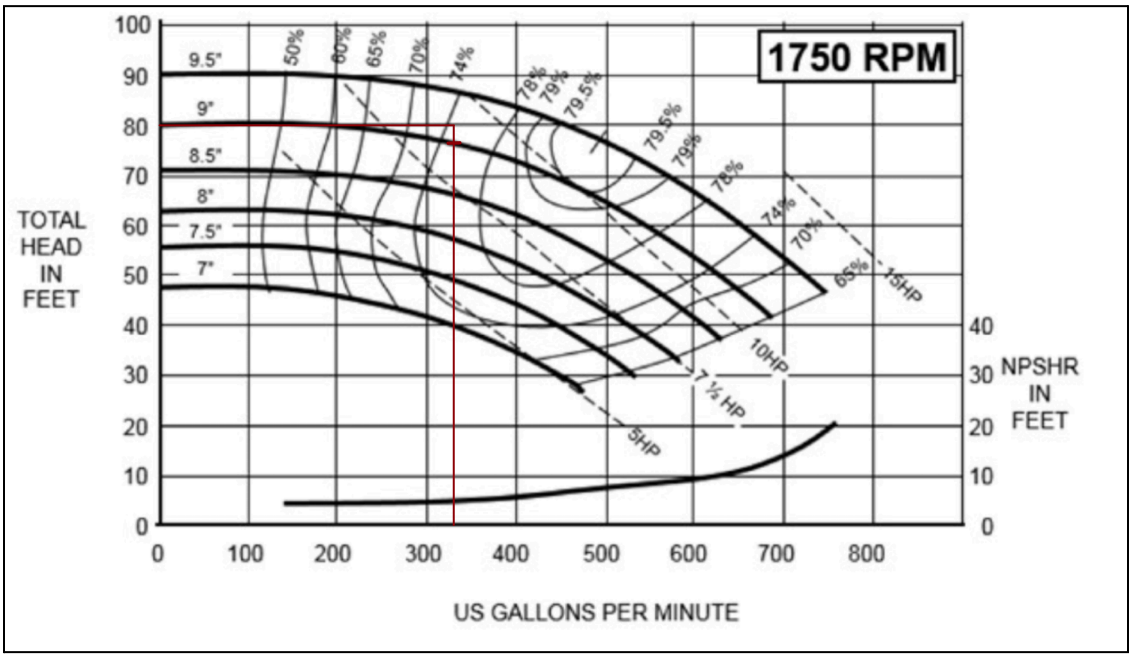


Figure 2: Pump Curve for Flow Rates of 0-800 GPM [3]

Using our final results, we compared them to various single stage pump curves that would provide a good pipe efficiency of above 70%. For our final result of 320-gpm with a rounded value of 80ft of head, we would need a 1750-rpm pump. The operating point falls closest to the 9 inch impeller curve, so the pump required would be a 1750-rpm centrifugal pump with a 9-in impeller.

Conclusions

In our design, we used PVC as our pipe material because it is cost effective, corrosion and chemically resistant, and easier to assemble because it is lightweight [1]. This makes PVC ideal for a rural location such as Waslala, Nicaragua because of limited resources and water quality. PVC is also durable and joints are resistant to leaking, which aids in water conservation [2]. In our code we included the roughness (ε) of copper to account for a pipe buildup of sediment or sludge over a long period of time, so our total required head was likely an overestimate for a PVC piping system. This accounts for instances if there is any inconsistency in pipe maintenance and cleaning out blockage in the school's pipe system. Given the final results, it was observed that the flow rate decreased with elevation, resulting in a very low flow rate for the top pipe (Q3). This is a poorly designed piping network as the ideal result would be to have the same flow rate throughout the pipe system. To improve our design, we would want to change diameters to manipulate head loss, specifically giving the pipe on top floor a wider diameter and the pipe on lower level a smaller diameter. With more major losses in pipe 1, changing the length would not be an efficient design choice because material would be wasted.

Appendix

Ai was used for the baseline of code and to implement the Colebrook-White Equation.

The following MATLAB script was used for our pipe design:

```
%% PROJECT 1 – SCRIPT

clear; clc;

global g rho mu gamma Q0 p0 Hreq ...
       D0 D1 D2 D3 D4 L0 L1 L2 L3 L4 ...
       A0 A1 A2 A3 A4 ...
       z0 z1 z2 z3 ...
       K1 K2 K3 ...
       eps_abs f0 f1 f2 f3 f4;
```

```
%% 1. CONSTANTS
g   = 9.81;     % m/s^2
rho = 1000;     % kg/m^3
mu  = 1.0e-3;  % Dynamic viscosity in Pa.s
gamma = rho*g; % N/m^3

Q0 = 0.02023;        % m^3/s  (given in project)
p0 = 15e3;        % Pa (15 kPa gage from project)
Hreq = 12;        % m required at each outlet

% GEOMETRY
z0 = 0;
z1 = 5;
z2 = 8;
z3 = 12; % z given in meters

D0 = 0.1016;   % m (~4 in)  main
D4 = 0.1016;   % m (~4 in)  riser
D1 = 0.0508;   % m (~2 in)  branch 1
D2 = 0.0508;   % m (~2 in)  branch 2
D3 = 0.1016;   % m (~4 in)  branch 3

L0 = 20;     % m  main
L4 = 10;     % m  riser (between bottom and top tees)
L1 = 15;     % m  branch 1
L2 = 15;     % m  branch 2
L3 = 15;     % m  branch 3

% Areas (m^2)
A0 = pi*D0^2/4;
A1 = pi*D1^2/4;
A2 = pi*D2^2/4;
A3 = pi*D3^2/4;
A4 = pi*D4^2/4;

% Inlet velocity (m/s)
V0 = Q0/A0;

% Copper roughness
eps_abs = 2.0e-6;    % micron

% K VALUES (flanged tees + elbows)
K1 = 1.75;
K2 = 1.95;
K3 = 1.35;
```

```
%% 2. First friction factor from main line (Colebrook)
Re0    = rho*V0*D0/mu;
epsrel = eps_abs/D0;
f0     = colebrook(Re0, epsrel);
```

```
% pipe friction factors
f1 = f0; f2 = f0; f3 = f0; f4 = f0;
```

```
%% 3. Solve for V1,V2,V3,V4,hp and iterate on f's

x = [1; 1; 1; 1; 10];                 % first guess [V1 V2 V3 V4 hp]^T
opts = optimset('Display','iter');

for iter = 1:4
    % Step 4: solve system for this set of f's
    x = fsolve(@eqns, x, opts);
    V1 = x(1); V2 = x(2); V3 = x(3); V4 = x(4); hp = x(5);

    % Step 5: update friction factors with new velocities
    Re1 = rho*V1*D1/mu;  f1 = colebrook(Re1, eps_abs/D1);
    Re2 = rho*V2*D2/mu;  f2 = colebrook(Re2, eps_abs/D2);
    Re3 = rho*V3*D3/mu;  f3 = colebrook(Re3, eps_abs/D3);
    Re4 = rho*V4*D4/mu;  f4 = colebrook(Re4, eps_abs/D4);

    fprintf('Iter %d: V1=%.3f  V2=%.3f  V3=%.3f  V4=%.3f  hp=%.3f\n',...
            iter,V1,V2,V3,V4,hp);
end
```

```
%% 4. Final flows and continuity check
Q1 = V1*A1; Q2 = V2*A2; Q3 = V3*A3; Q4 = V4*A4;

fprintf('\nFINAL RESULTS:\n');
fprintf('hp  = %.3f m of water\n', hp);
fprintf('Q0  = %.5f m^3/s\n', Q0);
fprintf('Q1 = %.5f,  Q2 = %.5f,  Q3 = %.5f m^3/s\n', Q1,Q2,Q3);

Qsum = Q1 + Q2 + Q3;
fprintf('Continuity: Q0 = %.5f,  Q1+Q2+Q3 = %.5f\n', Q0, Qsum);

% Convert m^3/s to gal/min
m3s_to_gpm = 15850.323;

Q0_gpm = Q0 * m3s_to_gpm;
Q1_gpm = Q1 * m3s_to_gpm;
Q2_gpm = Q2 * m3s_to_gpm;
Q3_gpm = Q3 * m3s_to_gpm;

fprintf('Q0 = %.2f gpm\n', Q0_gpm);
fprintf('Q1 = %.2f gpm, Q2 = %.2f gpm, Q3 = %.2f gpm\n', Q1_gpm, Q2_gpm, Q3_gpm);

% Convert head from meters to feet
m_to_ft = 3.28084;
hp_ft = hp * m_to_ft;

fprintf('hp = %.3f ft of water\n', hp_ft);
```

```
%% FUNCTIONS

function F = eqns(x)

    global g gamma Q0 p0 Hreq ...
           D0 D1 D2 D3 D4 L0 L1 L2 L3 L4 ...
           A0 A1 A2 A3 A4 ...
           z0 z1 z2 z3 ...
           K1 K2 K3 ...
           f0 f1 f2 f3 f4;
```

```
    V1 = x(1); V2 = x(2); V3 = x(3); V4 = x(4); hp = x(5);
    V0 = Q0/A0;

    % Flows (m^3/s)
    Q1 = V1*A1;
    Q2 = V2*A2;
    Q3 = V3*A3;
    Q4 = V4*A4;

    % 1) Continuity at bottom tee: Q0 = Q1 + Q4
    F1 = Q0 - (Q1 + Q4);

    % 2) Continuity at upper tee: Q4 = Q2 + Q3
    F2 = Q4 - (Q2 + Q3);

    % Major losses on each PATH (m)
    % Path to node 1: main + branch 1
    hL1_major = f0*(L0/D0)*(V0^2/(2*g)) + ...
                f1*(L1/D1)*(V1^2/(2*g));

    % Path to node 2: main + riser + branch 2
    hL2_major = f0*(L0/D0)*(V0^2/(2*g)) + ...
                f4*(L4/D4)*(V4^2/(2*g)) + ...
                f2*(L2/D2)*(V2^2/(2*g));

    % Path to node 3: main + riser + branch 3
    hL3_major = f0*(L0/D0)*(V0^2/(2*g)) + ...
                f4*(L4/D4)*(V4^2/(2*g)) + ...
                f3*(L3/D3)*(V3^2/(2*g));

    % Minor losses (m)
    hL1_minor = K1 * V1^2/(2*g);
    hL2_minor = K2 * V2^2/(2*g);
    hL3_minor = K3 * V3^2/(2*g);

    hL1 = hL1_major + hL1_minor;
    hL2 = hL2_major + hL2_minor;
    hL3 = hL3_major + hL3_minor;

    % 3–5) Bernoulli from inlet (0) to nodes 1,2,3
    F3 = z0 + p0/gamma + V0^2/(2*g) + hp ...
         - ( z1 + V1^2/(2*g) + hL1 + Hreq );

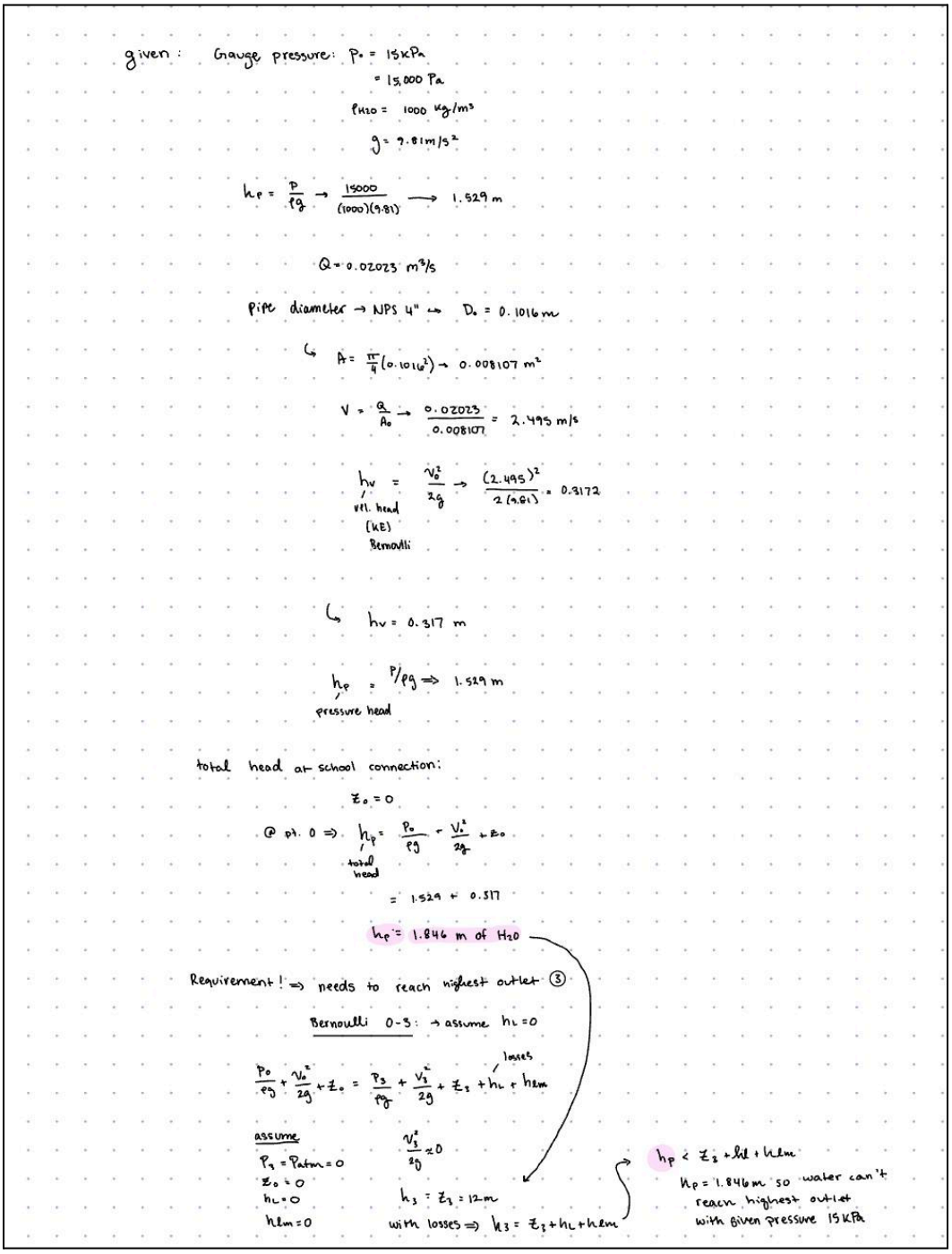
    F4 = z0 + p0/gamma + V0^2/(2*g) + hp ...
         - ( z2 + V2^2/(2*g) + hL2 + Hreq );

    F5 = z0 + p0/gamma + V0^2/(2*g) + hp ...
         - ( z3 + V3^2/(2*g) + hL3 + Hreq );

    F = [F1; F2; F3; F4; F5];
end

function f = colebrook(Re, eps_rel)
    % Colebrook–White equation for friction factor
    fun = @(f) 1./sqrt(f) + 2*log10( eps_rel/3.7 + 2.51./(Re.*sqrt(f)) );
    f0 = 0.02;
    f  = fzero(fun, f0);
end
```

Figure 1A: Calculations for Supply Condition of Water Line



References

[1] *Advantages of PVC Pipes in Building and Construction Industry*. NewTech. (2025, October 18). https://newtech-pipes.com/advantages-of-pvc-pipes/.

[2] *PVC pipe: Best choice for the environment*. PVC Pipe Association. (n.d.). https://www.uni-bell.org/About-Us/The-Environment.

[3] *Typical Single Stage Pump Curve*. https://www.pumpsandsystems.com/sites/default/files/article/2018-04-26%2010%3A12/pumpcurve.png. (n.d.).